Design Patterns - Lab 2
Proxy

Goal
Introduce the basic concept of the Proxy pattern and apply the Proxy pattern to our book.

Tasks
1. Start by adding a small delay to our Image constructor in order to simulate a load of a huge image.

```
Image(String name) {
	imageName = name;
	try {
		TimeUnit.SECONDS.sleep(5);
	} catch (InterruptedException e) {
		e.printStackTrace();
	}
}
```

2. Modify the main function to create 2 chapters with a number of images and print only the second chapter.

```
public static void main(String[] args) throws Exception {
	long startTime = System.currentTimeMillis();
	Image img1 = new Image("Pamela Anderson");
	Image img2 = new Image("Kim Kardashian");
	Image img3 = new Image("Kirby Griffin");
	Section playboyS1 = new Section("Front Cover");
	playboyS1.add(img1);
	Section playboyS2 = new Section("Summer Girls");
	playboyS2.add(img2);
	playboyS2.add(img3);
	Book playboy = new Book("Playboy");
```

```
      playboy.addContent(playboyS1);
      playboy.addContent(playboyS2);
      long endTime = System.currentTimeMillis();
      System.out.println("Creation of the content took " + (endTime -
startTime) + " milliseconds");

      startTime = System.currentTimeMillis();
      playboyS1.print();
      endTime = System.currentTimeMillis();

      System.out.println("Printing of the section 1 took " + (endTime -
startTime) + " milliseconds");

      startTime = System.currentTimeMillis();
      playboyS1.print();
      endTime = System.currentTimeMillis();

      System.out.println("Printing again the section 1 took " + (endTime -
startTime) + " milliseconds");
      }
```

3. What was the execution time for creating the content of the book: ……...

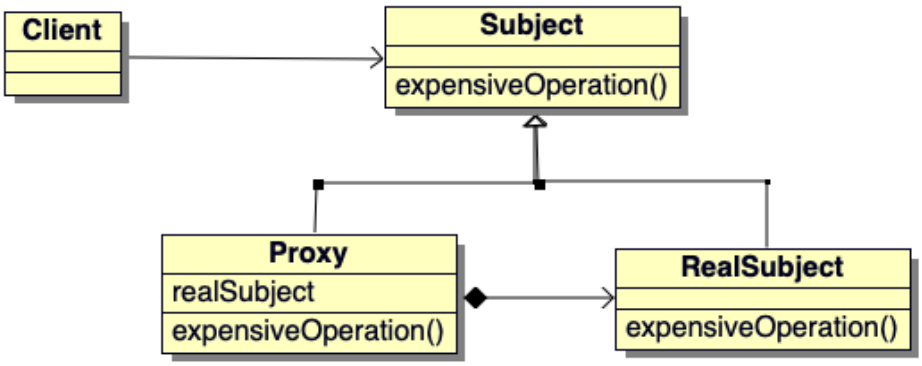
4. What was the execution time for printing the section 1 of the book: ……...

5. The section 1 has only one image and section 2 has two images would it be useful to load the images from section 2 only when we need them (at the print time)? Why? ……...

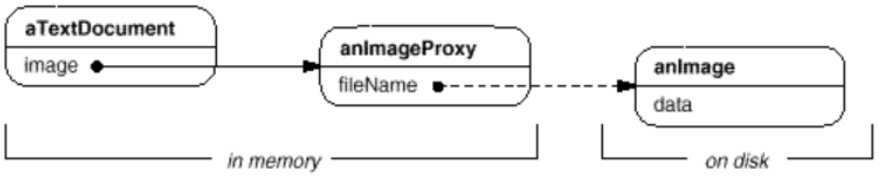
6. Do you have a solution for this problem? ……

7. What about proxy? ……...

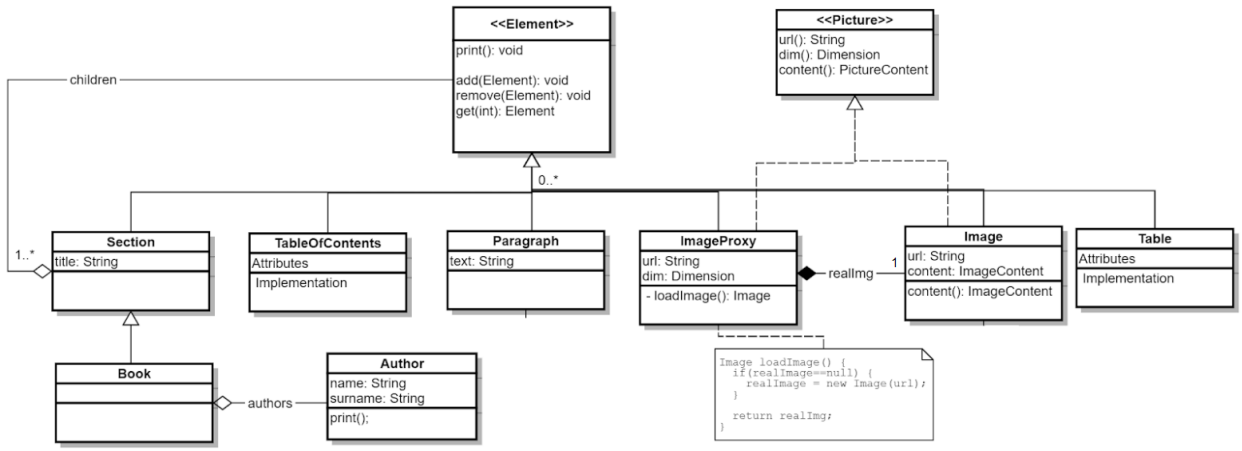
8. Proxy will help you delay the instantiation of the resource heavy object till the moment they are absolutely necessary. The structure of the pattern is very simple:



9. If we look to our image example the simplified view of the proxy is:



10. How should our class view look if we add an ImageProxy?



11. Types of proxy:
    a. Virtual proxy - placeholder for expensive operations
    b. Remote proxy - local representation of an object from a different space (stubs)
    c. Protective proxy - controls access

d. Smart proxy - smart pointers

12. Modify your project and add the missing classes and methods to implement the a ImageProxy and make sure the following main function is correctly executed.

```
public static void main(String[] args) throws Exception {
	long startTime = System.currentTimeMillis();
	ImageProxy img1 = new ImageProxy("Pamela Anderson");
	ImageProxy img2 = new ImageProxy("Kim Kardashian");
	ImageProxy img3 = new ImageProxy("Kirby Griffin");
	Section playboyS1 = new Section("Front Cover");
	playboyS1.add(img1);
	Section playboyS2 = new Section("Summer Girls");
	playboyS2.add(img2);
	playboyS2.add(img3);
	Book playboy = new Book("Playboy");
	playboy.addContent(playboyS1);
	playboy.addContent(playboyS2);
	long endTime = System.currentTimeMillis();
	System.out.println("Creation of the content took " + (endTime -
startTime) + " milliseconds");

	startTime = System.currentTimeMillis();
	playboyS1.print();
	endTime = System.currentTimeMillis();

	System.out.println("Printing of the section 1 took " + (endTime -
startTime) + " milliseconds");

	startTime = System.currentTimeMillis();
	playboyS1.print();
	endTime = System.currentTimeMillis();

	System.out.println("Printing again the section 1 took " + (endTime -
startTime) + " milliseconds");
}
```

13. What was the execution time for creating the content of the book after introducing the Proxy pattern:

……...

14. What was the execution time for printing the section 1 of the book:
    ……...
15. What was the total execution time of the main function? Are there any improvements vs the first implementation?
    ……...